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Quantitative data on urban flora inventories

Quantitative Assessment of Flora Inventories and Green Area Coverage in Urban Parks and Riverbanks of Estero Salado, Guayaquil

Quick Reference
Key Findings Table

Metric / Category	Estero Salado, Guayaquil (Direct Data)	Comparable Cities / Regions (for context)	Source(s)
% Introduced/Exotic Tree Species	Not available	24.8% (Curitiba), 50.9% (Recife)	1 2
% Native Terrestrial Tree Species	Not available	38.3% (Curitiba), 49.1% (Recife)	1 2
% Mangrove Tree Species (e.g., Rhizophora)	Not available	93% (Rhizophora, Indonesia, for context)	3
Green Area Coverage (m ² /inhabitant, Guayaquil)	Not available (see context below)	1.22 (Durán), 5–60 (Ecuadorian cities)	4 5
Green Area Coverage (total hectares, Guayaquil)	Not available	Not available	

Direct Answer

The current scientific literature does **not** provide explicit, recent quantitative figures on the percentages or counts of introduced/exotic, native terrestrial, or mangrove tree species, nor exact green area coverage metrics, specifically for urban parks and riverbanks of Estero Salado in Guayaquil. Available studies offer only indirect or comparative data from other cities and general trends in urban ecology, but no direct numbers with citations or DOIs addressing Estero Salado comprehensively were found.

Unable to Find

- No studies published since 2010 or recent inventories supply the exact percentages or counts for the three specified categories of tree species (introduced/exotic, native terrestrial, and mangrove species) specifically for Estero Salado's urban parks and riverbanks in Guayaquil, nor exact quantitative figures for green area coverage with accompanying DOIs.

Study Scope

- Time period:** Most relevant studies published post-2010; no recent (post-2015) direct inventories for Estero Salado.

- **Disciplines:** Urban ecology, botany, landscape planning, remote sensing.
- **Methods:** Systematic sampling, GIS mapping, UAV/LiDAR, mobile app-based inventories, literature review.

Assumptions & Limitations

- Data from comparable cities (e.g., Curitiba, Recife, Durán) are used for context but may not reflect Guayaquil's unique ecological and urban conditions.
- Absence of direct data for Estero Salado limits the ability to draw site-specific quantitative conclusions.
- Methodological diversity and lack of standardized species categorization hinder cross-study comparability.

Suggested Further Research

- Implement fine-scale, systematic flora inventories in Estero Salado using advanced remote sensing and GIS.
- Develop and publish updated, open-access datasets on urban green area coverage and species composition.
- Integrate citizen science and interdisciplinary approaches to improve data completeness and urban planning relevance.

1. Introduction

Urban green spaces are critical for biodiversity conservation, climate regulation, and human well-being, especially in rapidly urbanizing cities like Guayaquil. The Estero Salado, a prominent estuarine system traversing the city, and its adjacent urban parks and riverbanks, represent key ecological assets. Quantitative inventories of urban flora—distinguishing between introduced/exotic, native terrestrial, and mangrove species—are essential for informed urban ecological planning, restoration, and management. However, as this review demonstrates, there is a significant gap in site-specific, quantitative data for these categories in Guayaquil, particularly along the Estero Salado.

1.1 Context and Rationale

Estero Salado is both an ecological corridor and a socio-environmental landmark in Guayaquil, supporting diverse flora and fauna while facing intense urban pressures. The lack of comprehensive, quantitative flora inventories impedes effective conservation and urban planning, especially as the city continues to expand and transform its green infrastructure. Understanding the composition and coverage of native, exotic, and mangrove species is vital for maintaining ecosystem services, mitigating urban heat islands, and preserving local identity in the face of biotic homogenization and habitat loss [6](#).

2. Flora Composition in Urban Parks and Riverbanks of Estero Salado

2.1 Proportions of Introduced/Exotic, Native Terrestrial, and Mangrove Species

2.1.1 Available Data and Comparative Context

- **Estero Salado, Guayaquil:** No direct, recent quantitative inventories were found that provide the exact percentages or counts of tree species in the three requested categories for this area.
- **Comparable Cities:**

- **Curitiba, Brazil:** Urban parks inventory reported 24.8% exotic species, 38.3% native to Curitiba, and 24.8% native to Brazil, with 13 invasive alien species among the introduced trees [1](#).
- **Recife, Brazil:** Urban green spaces had 49.1% native and 50.9% exotic tree species; 37% of individual trees were native and 63% exotic. Notably, 86% of species did not naturally occur in the local phytogeographic domain, indicating a high prevalence of exotics [2](#).
- **Guayaquil Trends:** Studies highlight a dominant use of introduced (exotic) species in urban green spaces, leading to a loss of local vegetation identity and increased maintenance costs. Native species are underrepresented, and there is a call for their increased use to conserve local flora and mitigate urban heat island effects [6](#).

2.1.2 Synthesis

While the dominance of exotic species and underrepresentation of native flora is a common trend in Latin American urban parks, the absence of direct data for Estero Salado underscores a critical research gap. The available comparative data suggest that Guayaquil likely follows similar patterns, but this remains to be empirically validated.

2.2 Mangrove Species Abundance and Coverage

- **Estero Salado:** No recent quantitative inventory details the abundance or coverage of *Rhizophora* or other mangrove species specifically along Estero Salado's riverbanks [3](#) [7](#) [8](#).
- **Other Regions (for context):**
 - **Indonesia:** 93% of mangroves in a surveyed area were *Rhizophora* spp.
 - **Costa Rica:** *Rhizophora mangle* and *R. racemosa* dominate canal edges.
- **General Trends:** Mangrove ecosystems often show dominance by *Rhizophora* species, but the specific composition and abundance in Estero Salado remain undocumented in the scientific literature.

2.3 Spatial Patterns and Species Distribution

- **Guayaquil:** The use of imported vegetation has led to a homogenization of urban flora, with native species being spatially underrepresented. This trend is exacerbated by urban expansion and insufficient integration of native/endemic species in landscaping [6](#).
- **Spatial Data Gaps:** No studies provide detailed spatial distribution or mapping of species composition within Guayaquil's urban parks and riverbanks.

Synthesis

The lack of systematic, spatially explicit inventories for Estero Salado limits the understanding of species distribution patterns and the ecological integrity of urban green spaces in Guayaquil.

3. Green Area Coverage in Guayaquil

3.1 Quantitative Estimates of Green Area Coverage

- **Guayaquil (Direct Data):** No recent, exact figures for green area coverage (m^2 per inhabitant or total hectares) are available for the city proper.
- **Nearby/Comparable Cities:**
 - **Durán (Guayas province):** Urban green index is 1.22 m^2 per inhabitant, with some areas as low as $0.30 \text{ m}^2/\text{inhabitant}$ —well below WHO recommendations [4](#).
 - **Ecuadorian Cities (Urban Green Space Index, UGSI):** Cities with 50,000 inhabitants average $60 \text{ m}^2/\text{person}$; cities with 300,000 inhabitants (comparable to Guayaquil) have about $5 \text{ m}^2/\text{person}$ [5](#).
- **Guayaquil Trends:** Urban growth has led to a significant reduction in green spaces, with environmental impacts such as increased urban heat and loss of ecosystem services [6](#).

3.2 Comparative Analysis with Other Ecuadorian Cities

- **UGSI Data:** There is a strong inverse relationship between city size and per capita green space. Guayaquil, as Ecuador's largest city, likely has green space availability at or below $5 \text{ m}^2/\text{person}$, but no direct, recent figures are published [4](#) [5](#).
- **Durán:** Extremely low green space per capita, suggesting similar or slightly better conditions in Guayaquil's metropolitan area.

3.3 Spatial Distribution and Mapping of Green Areas

- **GIS/Spatial Data:** No recent studies provide neighborhood-level GIS-based quantifications or spatial distribution maps of green areas in Guayaquil, especially near Estero Salado [5](#) [9](#).
- **Environmental Value:** Estero Salado is recognized for its ecological importance, but detailed mapping and quantification of adjacent urban green spaces are lacking.

Synthesis

The absence of up-to-date, spatially explicit data on green area coverage in Guayaquil hinders effective urban planning and equitable access to green spaces.

4. Methodological Approaches in Flora Inventories

4.1 Sampling and Data Collection Methods

- **Systematic Strip Sampling:** Effective for quantifying tree abundance and basal area in riparian zones; likely applicable to Estero Salado [10](#) [11](#).
- **Stratified Random Plots:** Used in urban park inventories to assess community composition and calculate importance value indices [12](#) [13](#).
- **Mobile App-Based Inventories:** Expert systems for tree data collection in riparian areas have achieved $>90\%$ accuracy compared to manual methods [14](#).

- **UAV/LiDAR:** Advanced remote sensing provides accurate 3D structural data for tree inventories [15](#).
- **Vegetation Surveys & Multivariate Analysis:** Used to relate species composition to environmental gradients and park types [13](#) [16](#).

4.2 Temporal and Spatial Patterns in Flora Inventories

- **Temporal Patterns:** Seasonal and interannual variations in species abundance are linked to temperature, precipitation, and water conditions. Sampling timing significantly affects species detectability and richness [17](#) [18](#) [19](#).
- **Spatial Patterns:** Low floristic similarity between sectors due to habitat heterogeneity; strong vegetation gradients along river basins [13](#) [16](#).

Synthesis

Methodological diversity is a strength in urban flora inventories, but the lack of standardized protocols and comprehensive, long-term monitoring limits data comparability and utility for urban planning.

5. Data Gaps and Recommendations for Future Research

5.1 Current Data Gaps in Flora Inventories

- **Systematic Sampling Deficiency:** Most studies lack systematic, fine-scale sampling in Estero Salado and broader Guayaquil [6](#) [20](#).
- **Species Categorization:** Insufficient data on native vs. exotic vs. mangrove species, with a tendency to overlook functional and successional group classifications [20](#).
- **Spatial Coverage:** Inventories are spatially biased toward accessible or central urban areas, neglecting peripheral or less accessible zones [21](#).
- **Integration with Urban Planning:** Limited incorporation of scientific research and local participation in urban development processes [22](#) [23](#).

5.2 Recommendations for Improved Urban Flora Assessment

- **Adopt Advanced Technologies:** Use UAV, LiDAR, and GIS mapping for high-resolution, spatially explicit inventories [15](#) [24](#).
- **Promote Native/Endemic Species:** Integrate native flora into urban green space design to enhance biodiversity and ecosystem services [6](#).
- **Citizen Science & Stakeholder Engagement:** Involve local communities and use mobile applications to expand data collection and monitoring [20](#) [22](#).
- **Standardize Methodologies:** Develop and implement standardized protocols for species categorization and inventory design.

- **Interdisciplinary Approaches:** Combine ecological, urban planning, and social science perspectives for holistic urban green space management [25](#).

Synthesis

Bridging the data gaps in Guayaquil's urban flora inventories requires methodological innovation, stakeholder engagement, and a commitment to open, accessible data. These steps are essential for evidence-based urban ecological planning and the sustainable management of Estero Salado's unique green spaces.

6. Real-world Implications

- **Urban Planning:** Lack of quantitative data on species composition and green area coverage impedes effective urban planning, restoration, and climate adaptation strategies in Guayaquil.
- **Biodiversity Conservation:** Underrepresentation of native and mangrove species threatens local biodiversity and ecosystem resilience.
- **Social Equity:** Insufficient and unevenly distributed green spaces exacerbate socio-environmental inequalities in rapidly urbanizing areas.
- **Policy Development:** Evidence-based policies require robust, up-to-date data on urban flora and green space distribution.

7. Future Research Directions

- **Comprehensive Inventories:** Conduct systematic, fine-scale inventories of tree species in Estero Salado's urban parks and riverbanks, explicitly categorizing species as introduced/exotic, native terrestrial, or mangrove.
- **Spatial Mapping:** Develop GIS-based spatial distribution maps of green areas and species composition at the neighborhood level.
- **Longitudinal Monitoring:** Establish long-term monitoring programs to track changes in species composition and green area coverage.
- **Interdisciplinary Integration:** Foster collaboration between ecologists, urban planners, and local communities to ensure research informs practical urban management.
- **Open Data Initiatives:** Publish datasets and inventory results in open-access repositories to facilitate transparency and further research.

Synthesis:

The current state of knowledge on urban flora in Guayaquil, and specifically along Estero Salado, is characterized by significant data gaps, particularly in the quantitative categorization of tree species and green area coverage. While comparative data from other Latin American cities provide useful context, there is an urgent need for targeted, high-resolution research in Guayaquil. Addressing these gaps will support more

effective urban ecological planning, biodiversity conservation, and equitable access to green spaces in one of Ecuador's most dynamic urban environments.

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